

The Experiment and Simulation for Water Movement of Preferential Flow in Soil

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A large number of lab and field experiments have showed preferential flow is a ubiquitous phenomenon. Through analysis on the field dye experiment, precipitation and infiltration data in the Experimental Station in Zhengzhou and percolation test data in lab and field experiment, the paper analyze the characterization of preferential flow by a method of statistical analysis and discuss the form mechanism together with influencing factor in soil. The results show the field dye experiment of silt loam in Zhumadian, precipitation and infiltration data and the percolation test in the Experimental Station have showed the macropore caused by the root and wormhole greatly develops in the upper soil layer, tortuouse and decreases with the depth. However, the cranny caused by dry shrinkage comparatively develops in the deep soil layer, the macropore has good connectivity, and therefore preferential flow easily occurs in silt loam than sand silt in Kaifeng.

Keywords: Preferential Flow; Hydrus; Two-Region Model; Analytical Solution.

1. Introduction

A large number of lab and field experiments have showed preferential flow is a ubiquitous phenomenon [1]. Unlike micropores where water moves upward or downward based on hydraulic gradient, in macropores, water flows predominantly downward due to the gravity [2]. The occurrence of preferential flow in soil greatly reduces the availability of water and nutrient for the crops [3]. Preferential flow paths also have influence on podzols in coniferous forests [4]. Meanwhile, as a result of small area and short time contact with soil matrix, the remaining contamination such as fertilizer and pesticide in soil is not ready to decay and transfers to groundwater via preferential flow pathways, which is one cause of groundwater contamination. And Some evidences from field studies confirmed that 70–85% of water flux was ascribed to preferential flow, which is responsible for the difficulty in predicting water and solute movement in field condition [5,6]. Therefore, preferential flow has changed the original understanding of groundwater recharge system and the transport rule [7].

Through analysis on the field dye experiment, precipitation and infiltration data in the Experimental Station in Zhengzhou and percolation test data in lab and field experiment, the paper analyze the characterization of preferential flow by a method of statistical analysis and discuss the form mechanism together with influencing factor in soil. Through simulation on Hydrus in the lab percolation test, preferential flow transport in soil has been characterized by a qualitative–semiquantitative method.

2. Materials and Methods

The experimental area is located in the Wollonggong Village, Zhongyuan District, which is in the southwest of Zhengzhou. Zhengzhou is a warm temperate semi-arid climate and four distinctive seasons. The underground observing room of the lysimeter is a semi-underground bunker building, 8m in diameter and height 11.5 meters. The 35 sets of lysimeter are ring around the underground observing room. The soil sample collected from the plain area of Zhengzhou are divided into 7 groups. These seven groups of soil sample represent the lithologic character of the soil in the plain area of Zhengzhou. The fine sandy is disturbed sample, and the others are undisturbed samples. Every group is divided into 5 copies as the designed depth of water of 1m, 2m, 3m, 5m and 7m. There are 35 samples in total.

The meteorological field includes a set of rain gauge of 20cm in diameter and the Siphon-type Self-recording Rain Gauge, two sets of evaporating dish of 20cm and 61.8cm in diameter, a big louvered screen with recording thermometer and hygrometer in it, a small louvered screen with psychrometer and hygrometer in it, anemorumbograph, barometer, geothermometer, freezing meter and so on.

The two samples to 1m, 2m, 3m, 7m depth were basin irrigation artificially of 9g/L sodium chloride solution respectively, and washed by clear water under simulated rainfall in the experiment. In the experiments, continuous observation of NaCl solution infiltration amount of day and night in the underground observation room, record the volume of the flow liquid and measure the concentration of the flow liquid to get the relationship between the volume and concentration change with time of the bottom soil column interface under different depth and lithologic.

3. Results and Discussions

3.1. Analysis of the data

The research and the dyeing experiment in the field shows the preferential flow is common in the clay soil. There are many pore passages in the clay soil. When

the rainfall and the irrigation water quantity is too amount to be all absorbed by tiny pores in time, a part of the water would arrive in underground water infiltration rapidly through the macro pore with good permeability. So the water of infiltration recharges the underground water and does not have to add the all water deficit of the aeration zone. This is the so-called preferential flow.

The infiltration law of Zhumadian Clay under different depth is stable and the rainfall charge the underground water in a short time, but the next infiltration amount reduces as the depth is more deep. When the depth is less than 2m, the water of the rainfall infiltrate directly. The infiltration is over soon after the rainfall stops. However, when the depth is beyond 3m, after the peak of the infiltration there still exist continuous next infiltration with little amount. It means there are preferential flow supplies in the Zhumadian Clay mainly, and there exist some plug flow supplies in the clay when the water level is enough deep.

3.2. The result of the balancing field infiltration experiment and discussion

As the analysis of the data, we can find the infiltration flow always exist in the Zhumadian Mild Clay under different depth, and some plug flow also exist in it when the depth is enough deep. Qi find that in the Kaifeng Fine Sand the plug flow mainly exists, so the experiment make the contrast test on the Kaifeng Fine Sand to get the curve of the water penetration through the change of the tracer concentration in the flow fluid, expound the infiltration process of the preferential flow, and analysis the regularity and characteristics of the preferential flow.

In the experiment, the rule of the penetration in the Zhumadian Mild Clay and the Kaifeng Fine Sand changes under different depth. The preferential flow exist in all depth of the Zhumadian Mild Clay as the infiltration experiment. The sharp of the curve is single peak and symmetric except the 7m depth in the Kaifeng Find Sand, we can find there are mainly plug flow supplies in the Kaifeng Find Sand. As the concentration of the output tracer we can get the temporal variation laws of the chloride concentration in Zhumadian Mild Clay and Kaifeng Fine Sand. In the figure, the water breakthrough time in Zhumadian Mild Clay is obviously earlier than it in the Kaifeng Fine Sand, and the Zhumadian Mild Clay has higher concentration of the solution in the flow fluid than the Kaifeng Fine has Sand at the beginning. It comes to the conclusion that different soil-forming processes and geographical conditions make the different structures, so the connectivity of the macropore and the physic length of the macropore effected by the bending of it are different, that is the main reason

makes the concentration of the output solution differently. In the figure we can also find that in the other 3 soil column except the burial depth of 1m, the time reach the breakthrough peak and the concentration of the solution in the flow fluid are nearly the same, the time are 480minutes, 480minnutes, 663minutes and the concentrations are 7750, 5680, 5220 $\mu\text{s}/\text{cm}$ in the burial depth of 2m, 3m, 7m. The macropore in the Zhumadian Mild Clay is with good connectivity. The water recharge the underground water through the macropore channel as soon as the experiment starts, and the infiltration is over when the experiment stops. It just the difference of the breakthrough time made by the physic length of the macropore.

In Fig. 1, the groundwater recharge rate in the Zhumadian Mild Clay is much more at the beginning of the experiment because of the preferential flow supply exists in the Zhumadian Mild Clay. However, the groundwater recharge rate in the Zhumadian Mild Clay changes very little after increasing the simulated rainfall, and it is less in the later stage of the experiment. (Except in the soil column in the burial depth of 7m). It has close relationship to the soil granule composition. Zhumadian Mild Clay with clay content was 28.8%, compared to 4.7% of Kaifeng Fine Sand, Zhumadian Mild Clay content is 6 times of the later. By the late, due to the increase of soil water content, the clay expansion occurs, causing soil macropore plugging, the soil moisture is supplied mainly by the piston infiltration, changes in groundwater recharge is not obvious. And in the Kaifeng Fine Sand, groundwater recharge is given priority to with piston supplies; groundwater recharge is increasing until the end of the experiment.

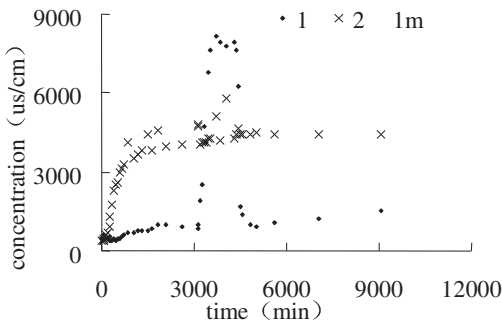


Fig. 1. The breakthrough curve of chloride ion under different depth.

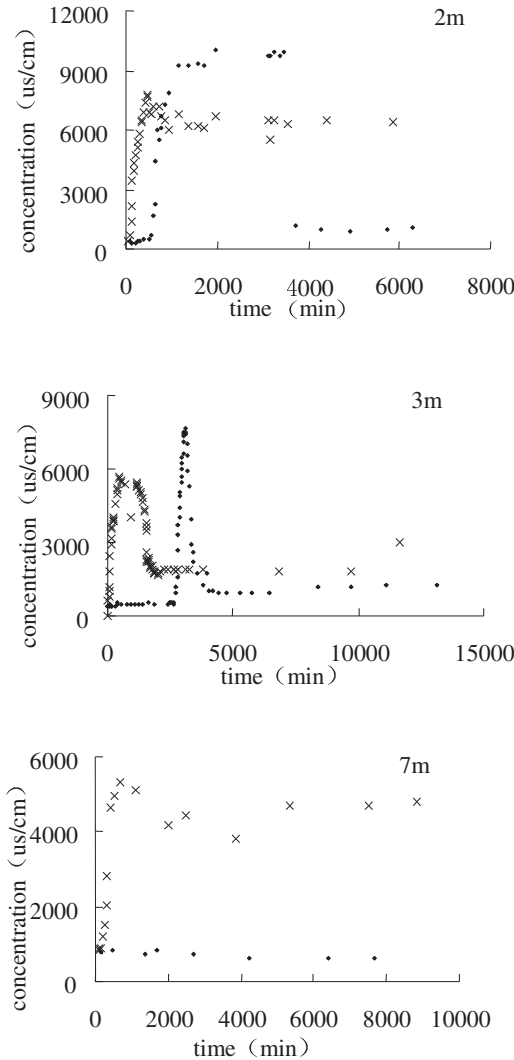


Fig. 1. (Continued).

4. Conclusion

The experiment shows the macropores are developed by root and the wormhole at the superficial part of the Zhumadian Mild Clay, and the macropores at the deep developed by crack formed by dry shrinkage of the clay. The macropores have great connectivity, what makes preferential flow easier to exist in it.

The preferential flow has the characteristic of rapid breakthrough, and There is a certain correlation between the breakthrough time and the height of the soil column. The skewness value of the breakthrough curve is 1.07 which is beyond 0, the curve is asymmetric and left.

The preferential flow is effected by soil particle composition, pore structure, initial soil moisture content, variation of soil water content, the continuity of the macropore, rainfall, the upper boundary condition and other factors. Generally, preferential flow is easier to be produced if the soil texture is fine and the initial soil water content is higher under the same water flux density. And there is no preferential flow if the precipitation is less than 20mm in the research.

We should consider all aspects of the factors comprehensively in the study of the formation mechanism of preferential flow. It should be noted that the effect of macropore on the preferential flow is very complex. There is no fixed rule of the effect of the macropore on the time, the void volume, the peak of the breakthrough. The progress cannot simply described by the increase or decrease in pore.

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References

1. Beven K, Germann P, 1982. Macropores and water flow in soils. *Water Resources Res* 18(5): 1311–25.
2. Duraisamy S. Saravanathiiban, M. Emin Kutay, Milind V. Khire, 2014. Effect of macropore tortuosity and morphology on preferential flow through saturated soil: A Lattice Boltzmann study. *Computers and Geotechnics* 59, 44–53.
3. Christina Bogner, Werner Borken, Bernd Huwe, 2012. Impact of preferential flow on soil chemistry of a podzol. *Geoderma* 175–176, 37–46.
4. Z.B. Zhang, X. Peng, H. Zhou, H. Lin, X. Sun, 2014. Characteristics of cracks in two paddy soils and their impacts on preferential flow. *Geoderma* 228–229, 114–121.
5. Watson, K., Luxmoore, R., 1986. Estimating macroporosity in a forest watershed by use of a tension infiltrometer. *Soil Science Society of America Journal* 50, 578–582.

6. Wilson, G., Luxmoore, R., 1988. Infiltration, macroporosity, and mesoporosity distributions on two forested watersheds. *Soil Science Society of America Journal* 52, 329–335.
7. Z.B. Zhang, X. Peng, H. Zhou, H. Lin, H. Sun, 2015. Characterizing preferential flow in cracked paddy soils using computed tomography and breakthrough curve. *Soil & Tillage Research Journal* 146, 53–65.